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GNIOT Institute of Management Studies – PGDM (Batch 2021-23)

(TRIMESTER – IV) END TERM EXAMINATION, SEPTEMBER - 2022

FINANCIAL DERIVATIVES

[Time: 2 Hours 30 Minutes]

[Maximum Marks: 100]

INSTRUCTIONS:

- (i) **There are THREE sections in this question paper.**
- (ii) **Attempt questions from each section as per the directions.**
- (iii) **Make suitable assumptions wherever necessary.**

SECTION-A

(Short Answer Type Questions, do not exceed 100 words)

Q.1. Attempt all parts of the following:

(10×3=30)

Compare the following in your own words:

- (a) Exchange Traded vs Over the Counter Contracts
- (b) American Option vs European Options
- (c) Cash Settlement vs Physical Settlement of Futures Contract
- (d) Speculator vs Investor
- (e) Forward commitments vs Contingent Claims as a category of derivatives
- (f) Interest Rate vs Currency Swaps
- (g) Future vs Cash Market
- (h) Fixed Interest Rate vs Floating Interest Rate
- (i) Pricing vs Valuation of Swaps
- (j) Stock Trading vs Option Trading

SECTION-B

(Long Answer Type Questions)

Attempt all questions of the following:

(4×10=40)

Q.2. (a) Explain with example the concept of in the money and out of the money call option?

OR

(b) Explain with example the concept of in the money and out of the money put option?

Q.3. (a) What do you mean by Forward Contract? List down the features of forward contract?

OR

(b) What do you mean by Future Contract? List down the features of forward contract?

Q.4. (a) Explain the concept of Cost of Carry Model? Discuss the Cash and Carry Trade arbitrage strategy.

OR

(b) Explain the concept of Cost of Carry Model? Discuss the Reverse Cash and Carry Trade arbitrage strategy.

Q.5. (a) Justify the following statements:

- 1) Delta is positively related to call options.
- 2) The closer to expiration the option is, the smaller the extrinsic value.

OR

(b) Justify the following statements:

- 1) Delta is negatively related to put options.
- 2) Theta option Greek is also referred to as time decay.

SECTION-C

Q.6. Attempt all parts of the following:

(2×15=30)

Mr. A is closely tracking the movement in XYZ Ltd. stock. The stock during the past 3 months has fallen from a high of Rs. 140 to Rs. 100. However, Mr. A now believes that the stock will bounce back from this level of Rs. 100 and is expecting a jump of 15% from this level in next one month.

Keeping this in mind Mr. A went long on a call option on XYZ Ltd. stock with one month expiry; strike price of Rs. 100. Mr. A paid a premium of Rs. 5 for the call option.

Mr. B, who is also closely tracking the movement of XYZ stock has a bearish view on the stock and is expecting price to fall further by 10% from the current level of Rs. 100 in next one month. Hence, Mr. B is planning to go long on a put option on XYZ stock with one month expiry; strike price Rs. 100. Mr. B paid a premium of Rs. 10 for the put option.

One option lot size is equal to 1 share of XYZ Ltd.

- 6. (a)** Hypothetically assuming that if Mr. A is right in his analysis and the stock price at the end of expiry is exactly at the level as he expected, calculate the payoff to Mr. A on expiry. Also calculate the payoff to short call in this scenario (provide proper explanation along with calculations). Prepare a general payoff diagram for long call and short call and provide a general explanation of the same.
- 6. (b)** Hypothetically assuming that if Mr. B is right in his analysis and the stock price at the end of expiry is exactly at the level as he expected, calculate the payoff to Mr. B on expiry. Also calculate the payoff to short put in this scenario (provide proper explanation along with calculations). Prepare a general payoff diagram for long put and short put strategy and provide a general explanation of the same.